- Nucleotides like cAMP (cyclic AMP) and cGMP (cyclic GMP) act as second messengers involved in hormonal signaling pathways.

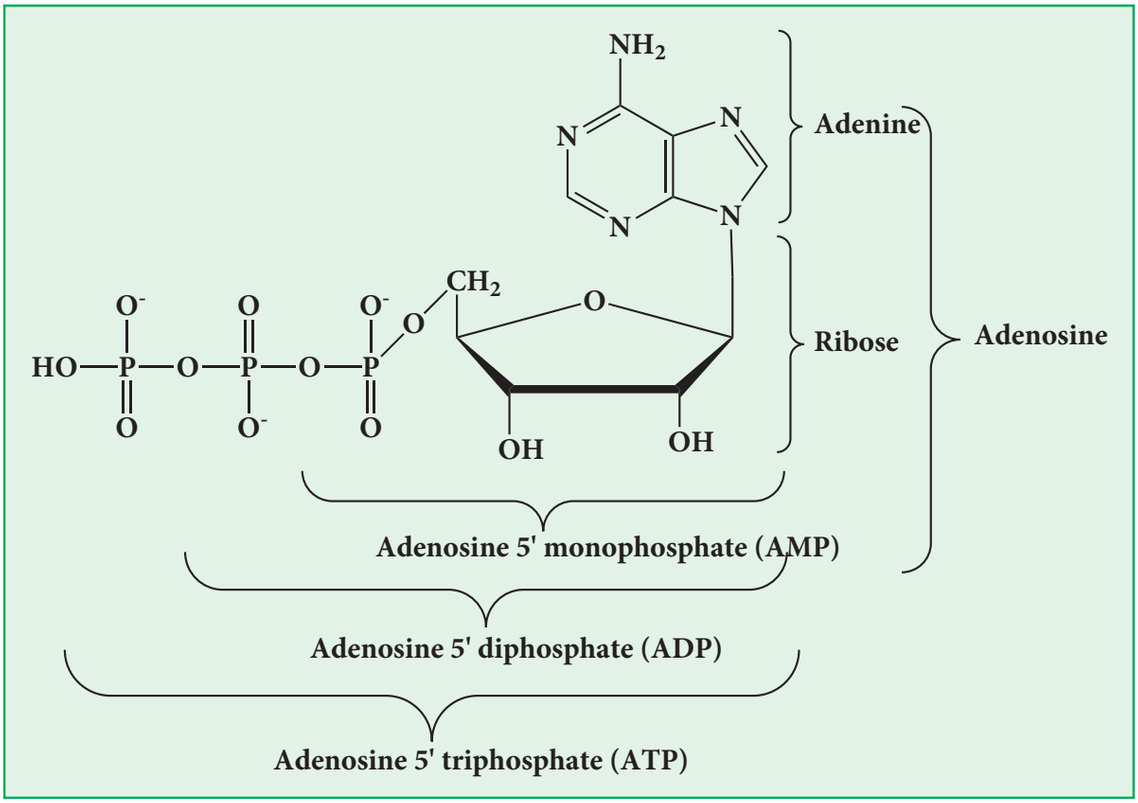

Figure 7.5 Structure of a nucleotide

DO YOU KNOW?

Nucleosides generally do not have physiological role. However, adenosine is an exemption with biological activity. It acts as a local hormone and is involved in many different biological effects like blood vessel dilation. Supraventricular tachycardia, a heart condition characterized by rapid heart beat is treated with intravenous injections of adenosine. Adenosine promotes sleepiness. Caffeine, a chemical found in tea and coffee brings wakefulness by blocking the binding of adenosine to its receptors.

Oligonucleotides are polymers which yield two to ten residues of mononucleotides on hydrolysis. Two nucleotides join together to form dinucleotides. Example for biologically important dinucleotides are NAD and FAD, which act as co-enzymes.

Polynucleotides are polymers that yield more than ten nucleotides on hydrolysis. Polynucleotides have directions. If the first nucleotide of the sequence has 5′ triphosphate free and 3′OH group bonded to the next nucleotide, then the direction of the polynucleotide is 5′to 3′. If the first nucleotide in the sequence has 3′OH group free, then it is said to be from 3′to 5′direction.

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